

SAMAY THAKUR

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EDUCATION

M.S. Ramaiah Institute of Technology

Bachelor of Engineering (B.E.) | Electronics and Instrumentation Engineering | CGPA: 7.95

Oct 2023 – Present

Bengaluru, India

Sunbeam School Varuna

Grade X (CBSE): 89.6% | Grade XII (CBSE): 74.2%

Mar 2019 – Jun 2022

Varanasi, India

EXPERIENCE

CPS & IoT Intern

Aug 2024 – Aug 2024

Intra Institutional Internship, MSRIT

Bengaluru, India

- Designed a real-time water-level monitoring and automatic pump-control system on Arduino Uno to replace manual tank filling, using an ultrasonic sensor for non-contact water-level measurement.
- Implemented upper/lower threshold control logic in C++ to automatically switch the pump ON/OFF, preventing tank overflow and dry-run conditions.
- Validated the system across edge-case water levels, reducing manual intervention and verifying reliable automated pump control.

PROJECTS

CynLr EPM PoC | GitHub

- Structured a complex robotics PoC covering moving-object tracking and track-and-grasp into an executable program plan spanning robotics, CV/ML, software, mechanical, electronics, and external vendors.
- Mapped 41 workstreams and 97 subtasks in Asana, including task dependencies, intra-project dependencies, external dependencies, milestones, and risks.
- Developed a 604-person-hour effort model in Excel to allocate skill hours across engineering disciplines.
- Defined a KPI framework with lead/lag indicators across 5 dimensions—dependency resolution, development progress, infrastructure readiness, testing outcomes, and delivery readiness—for objective milestone tracking.
- Tools/Technologies: Asana, Microsoft Excel

SmartQC: On-Device Visual Quality Inspection | GitHub

Qualcomm | Snapdragon® AI Lab Build & Present Challenge

- Authored a PRD for small food producers lacking access to industrial inspection hardware, defining a field-accuracy target of at least $\geq 97\%$ field-accuracy target and documenting scope decisions before implementation.
- Engineered an on-device fresh-vs-mold bread classifier optimized through Qualcomm AI Hub for Snapdragon, achieving 99.2% validation accuracy (0.9997 ROC-AUC) with local NPU inference and no cloud calls.
- Evaluated the classifier on 352 uncured real-world images, achieving 97.4% field accuracy; used the validation-to-field performance gap to prioritize confidence-threshold flagging for low-confidence predictions.
- Developed a PyQt5 desktop application integrating explainability, batch processing, CPU-vs-NPU latency benchmarking, and SQLite-based inspection history.
- Tools/Technologies: Python, PyTorch, ONNX, Qualcomm AI Hub, PyQt5, OpenCV, Scikit-learn

AI-Based Bread Quality Indicator System | GitHub

- Designed a dual-modality bread-quality system combining MQ-135 gas sensing and ESP32-CAM vision to reduce reliance on a single detection modality.
- Built a MobileNetV2 classifier with 128×128 input, achieving 100% validation accuracy over 15 epochs and deploying inference on-device with TensorFlow Lite.
- Integrated Wi-Fi sensor telemetry, LED/buzzer alerts, and model predictions into a Flask dashboard for real-time quality alerts.
- Tools/Technologies: Python, TensorFlow, MobileNetV2, TensorFlow Lite, Flask, ESP32-CAM

Stock Market Price Prediction & Analysis | GitHub

- Analyzed 2019–2024 NIFTY 50 OHLCV data, framing next-day price forecasting as a time-series regression problem.
- Engineered technical-indicator features (MA20/MA50, RSI, MACD, Bollinger Bands) and trained a TensorFlow/Keras LSTM on 60-day sliding windows.
- Achieved 1.39% MAPE, 0.93 R^2 , and 71.8% directional accuracy, with MA20 and MA50 identified as the most influential features.
- Tools/Technologies: Python, TensorFlow, LSTM, Pandas, NumPy, Scikit-learn

TECHNICAL SKILLS

- Product & Program Management:** Asana, Microsoft Excel, Program Planning, Resource Planning, Dependency Management, Milestone Tracking, KPI Tracking, Risk Identification, Feature Prioritization, PRD Authoring, Cross-Functional Planning
- AI/ML:** PyTorch, TensorFlow, TensorFlow Lite, Scikit-learn, Computer Vision, Deep Learning, ONNX, Qualcomm AI Hub
- Data & Analytics:** SQL, Excel, Tableau, Power BI
- Engineering:** Python, C++, OpenCV, Flask, PyQt5, SQLite, Git, GitHub, VS Code, Jupyter
- Concepts:** Algorithms, Data Structures, DBMS, Operating Systems, Computer Networks, Digital Logic Design

ACHIEVEMENTS & LEADERSHIP

- Student Member:** Instrument Society of India (ISOI), IISc Bangalore (Nov 2023 – Oct 2027)
- Community Volunteer:** Organized and supported blood-donation drives through school and community initiatives, facilitating donor engagement and community welfare.